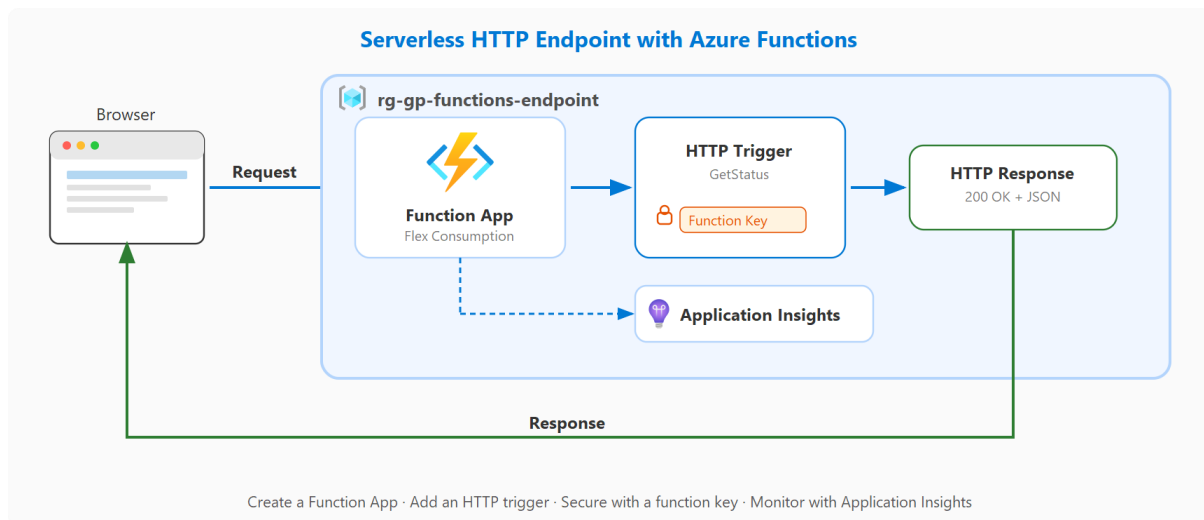
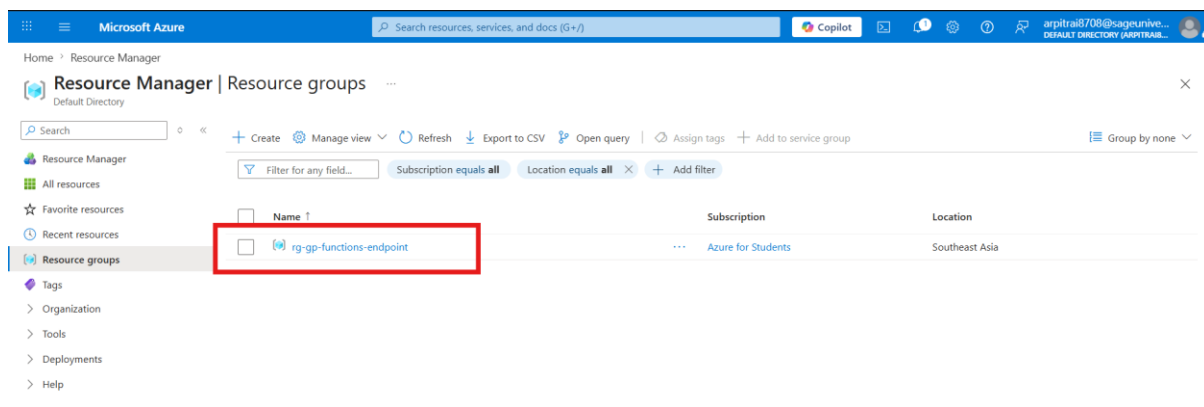


Project 1: Serverless HTTP Endpoint with Azure Functions



Exercise - Create the function app

Task 1: Prepare the environment



Task 2: Configure the Function App

The screenshot shows the Azure Portal interface for a Function App named 'func-gp-endpoint18'. The left sidebar contains navigation options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), Log stream, AI (preview), Resource visualizer, Functions, Deployment, Settings, Performance, Development Tools, API, and Monitoring. The main area displays the 'Essentials' tab with the following details:

- Resource group (move): rg-gp-functions-endpoint
- Status (move): Running
- Location (move): Southeast Asia
- Subscription (move): Azure for Students
- Subscription ID: 05ff0e42-2...
- Tags (edit): Add tags
- Default domain: func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurew...
- Operating System: Linux
- Plan type: Flex Consumption
- Instance Memory: 2048 MB

Below the essentials, there are tabs for Functions, Properties, and Notifications (0). A section titled 'Create functions in your preferred environment' offers two options:

- VS Code Desktop**: Best optimized for local development within VS Code, custom development tool requirements. Button: 'Create with VS Code Desktop'.
- Other editors or CLI**: Best optimized for using preferred editor for development, Visual Studio, IntelliJ, command line. Button: 'Set up your editor'.

Exercise - Create an HTTP-trigger function

Task 1: Open Cloud Shell

The screenshot shows the Azure Portal interface with the Cloud Shell open. The top bar includes the Microsoft Azure logo, a search bar, and user information. The main area displays the 'Overview' tab for the Function App 'func-gp-endpoint18'. The Cloud Shell terminal is active, showing the following text:

```
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
arpit [ ~ ]$
```

Task 2: Create the function project

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Overview' page for the 'func-gp-endpoint18' Function App. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), and Tags. The main content area is divided into two sections: 'Essentials' and 'JSON View'. The 'Essentials' section shows the Resource group (rg-gp-functions-endpoint), Status (Running), Location (Southeast Asia), and Subscription (Azure for Students). The 'JSON View' section shows the Default domain, Operating System (Linux), Plan type (Flex Consumption), and Instance Memory (2048 MB). Below the overview, there is a terminal window titled 'Welcome to Azure Cloud Shell'. The terminal output shows the following commands and their results:

```
Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
arpit [ ~ ]$ mkdir func-gp-endpoint && cd func-gp-endpoint
arpit [ ~/func-gp-endpoint ]$ func init --worker-runtime node --language javascript --model V4

The new Node.js programming model is generally available. Learn more at https://aka.ms/AzFuncNodeV4
Writing package.json
Writing .funcignore
Writing .gitignore
Writing host.json
Writing local.settings.json
Writing /home/arpit/func-gp-endpoint/.vscode/extensions.json
Running 'npm install'...
```

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Overview' page for the 'func-gp-endpoint18' Function App. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), and Tags. The main content area is divided into two sections: 'Essentials' and 'JSON View'. The 'Essentials' section shows the Resource group (rg-gp-functions-endpoint), Status (Running), Location (Southeast Asia), and Subscription (Azure for Students). The 'JSON View' section shows the Default domain, Operating System (Linux), Plan type (Flex Consumption), and Instance Memory (2048 MB). Below the overview, there is a terminal window titled 'Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.' The terminal output shows the following commands and their results:

```
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
arpit [ ~ ]$ mkdir func-gp-endpoint && cd func-gp-endpoint
arpit [ ~/func-gp-endpoint ]$ func init --worker-runtime node --language javascript --model V4

The new Node.js programming model is generally available. Learn more at https://aka.ms/AzFuncNodeV4
Writing package.json
Writing .funcignore
Writing .gitignore
Writing host.json
Writing local.settings.json
Writing /home/arpit/func-gp-endpoint/.vscode/extensions.json
Running 'npm install'...arpit func new --name GetStatus --template "HTTP trigger" --authlevel anonymous --authlevel anonymous
Select a number for template: HTTP trigger
Function name: [httpTrigger] The function "GetStatus" was created successfully from the "HTTP trigger" template.
arpit [ ~/func-gp-endpoint ]$ ls src/functions/
GetStatus.js
arpit [ ~/func-gp-endpoint ]$
```

Task 3: Deploy the function to Azure

The screenshot displays the Azure Portal interface for a function app named 'func-gp-endpoint18'. The 'Overview' tab is selected, showing the app's status as 'Running' and its location as 'Southeast Asia'. The 'Essentials' section lists the resource group, status, location, and subscription. The 'Default domain' is 'func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net'. The 'Operating System' is 'Linux', the 'Plan type' is 'Flex Consumption', and the 'Instance Memory' is '2048 MB'. The 'Activity log' section shows the deployment process, including the 'Kudu-SyncTriggerStep' starting, 'Kudu-CleanUpStep' starting and completing, and the 'Kudu-SyncTriggerStep' completing. The 'Host status' is 'Running', and the 'FunctionAppContentEditingState' is 'NotAllowed'. The 'extensionBundle' is 'Microsoft.Azure.Functions.ExtensionBundle, version: 4.35.0'. The 'GetStatus' command is shown, and the 'Invoke url' is 'https://func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net/api/getstatus'. The 'arpit' command is also shown.

Exercise - Test endpoint and review logs

Task 1: Test the HTTP endpoint in a browser

The screenshot shows a web browser window with the address bar displaying 'func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net/api/getstatus'. The page content displays 'Hello, world!'.

Task 2: Verify the function in the portal

Home > Function App > func-gp-endpoint18

Search resources, services, and docs (G+)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Log stream

AI (preview)

Resource visualizer

Functions

Deployment

Settings

Performance

Development Tools

API

Monitoring

Essentials

Resource group (move) : rg-gp-functions-endpoint

Status : Running

Location (move) : Southeast Asia

Subscription (move) : Azure for Students

Subscription ID : 05ff0...

Default domain : func-gp-endpoint18-hueSguayg8eadtb0.southeastasia-01.azure...

Operating System : Linux

Plan type : Flex Consumption

Instance Memory : 2048 MB

Tags (edit) : Add tags

Functions

Properties

Notifications (0)

Set up local environment

Refresh

Filter by name...

Name	Trigger	Status	Monitor
GetStatus	HTTP	Enabled	Invocations and more

Task 3: Enable Application Insights

Home > Function App > func-gp-endpoint18 > GetStatus

GetStatus | Invocations

func-gp-endpoint18

Code + Test

Integration

Function Keys

Invocations

Logs

Metrics

Open in Application Insights

Refresh

Send us your feedback

Query

Up to 20 of the most recent function invocation traces. For more advanced analysis, run the query in Application Insights.

Success count

7

Last 30 days

Error count

0

Last 30 days

Search

Date	Status	Result Code	Duration (ms)	Operation ID
6/8/2026, 12:30:16 PM	Success	200	5	31554df9439cdf2551d6b5bf43a141f
6/8/2026, 12:30:13 PM	Success	200	8	d8cbab36d4095ad07b895e634928fd02
6/7/2026, 5:52:57 PM	Success	200	6	cd0485329390dd2bcf44cee2261cbb6b
6/7/2026, 5:52:27 PM	Success	200	38	b76f97e086cedc3c8873ed3197aa00b
6/7/2026, 5:33:52 PM	Success	200	5	a31940b3302bd49dcf30c8ffa63a2150

Home > Function App > func-gp-endpoint18 > GetStatus | Invocations

Logs

func-gp-endpoint18

New Query 1*

New Query 2*

New Query 3*

Observability agent

Save

Share

User Query

Time range: Set in query

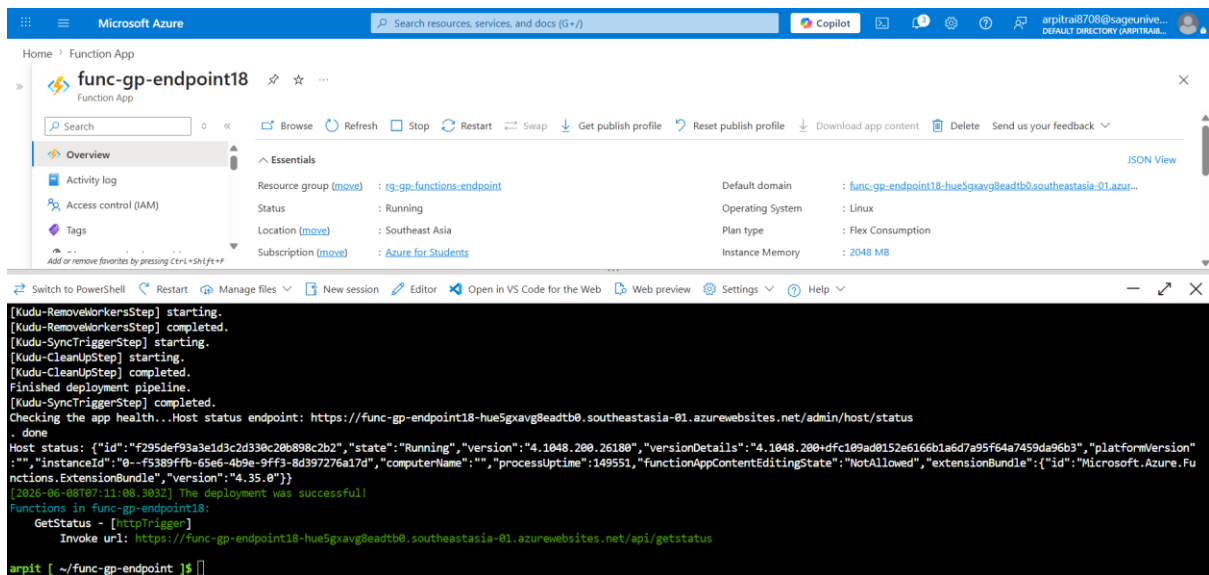
Show: 500000 results

Add

Results

timestamp [UTC]	id	operation_Name	success	resultCode	duration	operation_id	cloud_RoleName	invocationId
> 6/8/2026, 7:00:16.445 AM	74b03c7b1951fbd1	GetStatus	True	200	5.3818	31554df9439cdf2551d6b5bf43a141f	func-gp-endpoint18	dded3e1e-eee4-45c6-8027-1f953ce1
> 6/8/2026, 7:00:13.105 AM	a0fcf847e9e749fb	GetStatus	True	200	7.5853	d8cbab36d4095ad07b895e634928fd02	func-gp-endpoint18	f9d3f17e-e05f-44de-a396-efe25bcbt
> 6/7/2026, 12:22:57.355 PM	b4cf4d6adc59cab2	GetStatus	True	200	6.1135	cd0485329390dd2bcf44cee2261cbb6b	func-gp-endpoint18	1ac1264f-d907-4c4c-a06d-f6d22c58f
> 6/7/2026, 12:22:27.384 PM	06b0d0fb49ebfff	GetStatus	True	200	37.7838	b76f97e086cedc3c8873ed3197aa00b	func-gp-endpoint18	deb82238-492f-46d8-8621-a417120

Task 4: Restrict access to the function



Microsoft Azure

Search resources, services, and docs (G+)

Copilot

arpit8708@sageunive...
DEFAULT DIRECTORY (ARPT8708)

Home > Function App

func-gp-endpoint18
Function App

Search

Overview

Activity log

Access control (IAM)

Tags

Essentials

Resource group (move) : rg-gp-functions-endpoint

Status : Running

Location (move) : Southeast Asia

Subscription (move) : Azure for Students

Default domain : func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net

Operating System : Linux

Plan type : Flex Consumption

Instance Memory : 2048 MB

Switch to PowerShell

Restart

Manage files

New session

Editor

Open in VS Code for the Web

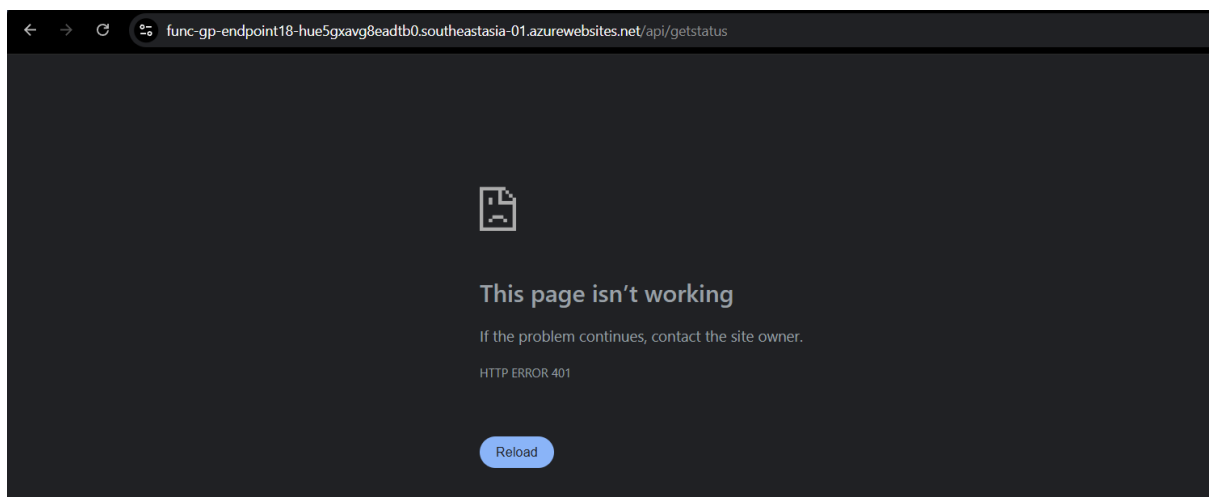
Web preview

Settings

Help

```
[Kudu-RemoveWorkersStep] starting.
[Kudu-RemoveWorkersStep] completed.
[Kudu-SyncTriggerStep] starting.
[Kudu-SyncTriggerStep] completed.
[Kudu-CleanUpStep] starting.
[Kudu-CleanUpStep] completed.
Finished deployment pipeline.
[Kudu-SyncTriggerStep] completed.
Checking the app health...Host status endpoint: https://func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net/admin/host/status
. done
Host status: {"id":"f295def93a3e1d3c2d330c20b898c2b2","state":"Running","version":"4.1048.200.26180","versionDetails":{"4.1048.200+dfc109ad8152e6166b1a6d7a95f64a7459da96b3","platformVersion":"","instanceId":"0--f5389ffb-65e6-4b9e-9ff3-8d397276a17d","computerName":"","processUptime":149551,"functionAppContentEditingState":"NotAllowed","extensionBundle":{"id":"Microsoft.Azure.Functions.ExtensionBundle","version":"4.35.0"}}}
[2025-06-08T07:11:08.303Z] The deployment was successful!
Functions in func-gp-endpoint18:
  GetStatus - [httpTrigger]
    Invoke url: https://func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net/api/getstatus
arpit [ ~/func-gp-endpoint ]$
```

Task 5: Test restricted access



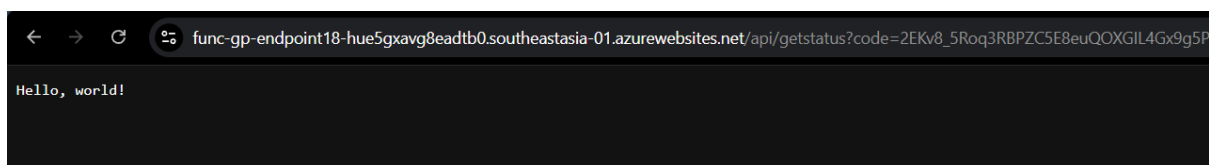
func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net/api/getstatus

This page isn't working

If the problem continues, contact the site owner.

HTTP ERROR 401

Reload



func-gp-endpoint18-hue5gxavg8eadtb0.southeastasia-01.azurewebsites.net/api/getstatus?code=2EKv8_5Roq3RBPZC5E8uQOXGIL4Gx9g5P

Hello, world!

Task 6: Review invocation logs

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

arpitrai8708@sageunive...
DEFAULT DIRECTORY (ARPTRAI...

Home > Function App > func-gp-endpoint18 > GetStatus

GetStatus | Invocations ...

func-gp-endpoint18

Code + TestIntegrationFunction KeysInvocationsLogsMetrics

Open in Application InsightsRefreshSend us your feedback

Up to 20 of the most recent function invocation traces. For more advanced analysis, run the query in Application Insights.

Success count

8

Last 30 days

Error count

0

Last 30 days

Search

Date	Status	Result Code	Duration (ms)	Operation ID
6/8/2026, 12:44:47 PM	Success	200	45	076211b946a7d99adb367c72966a7d0a
6/8/2026, 12:30:16 PM	Success	200	5	31554df9439fcd2551d6b5b43a141f
6/8/2026, 12:30:13 PM	Success	200	8	d8cbab36d4095ad07b895e634928fd02
6/7/2026, 5:52:57 PM	Success	200	6	cd0485329390dd2bcf44cee2261cbb6b
6/7/2026, 5:52:27 PM	Success	200	38	b76f97e086ce6dc3c8873ed3197aa00b